

When using AI assisted tools to generate initial prototypes, "Requirements lost in AI conversations" and "Prototype → engineering handoff gap" are most commonly discussing issues. Is this true ? Need evidences

Yes there is growing evidence that two of the most commonly reported problems in AI-assisted prototyping workflows are:

1. **Requirements getting lost, distorted, or fragmented across long AI conversations**
2. **A gap between AI-generated prototypes and production-ready engineering handoff**

1. Evidence for “Requirements lost in AI conversations”

Evidence A — Replit PM Guide

[Replit: A Product Manager’s Guide to Using AI to Build Working Prototypes](#)

Replit explicitly states:

“Every handoff to design, engineering, and QA introduces translation loss...”

This is strong evidence that requirements degrade through AI-assisted workflows and conversational prototyping.

The article also argues that behavioral briefs and structured specs reduce this loss.

Evidence B — Requirements Engineering Research

Academic literature also supports this concern.

Systematic Mapping Study

[Requirements Engineering for Artificial Intelligence Systems \(arXiv\)](#)

Researchers found:

- existing requirements engineering methods are insufficient for AI systems
- current tooling struggles with explainability, ambiguity, and evolving requirements

Industry Practices Study

[Requirements Practices and Gaps When Engineering Human-Centered AI Systems \(arXiv\)](#)

The survey found:

- many teams still rely on informal tools like Office docs/UML
- current tools are “not equipped to manage AI-based software”

- requirements quality suffers in AI workflows

2. Evidence for “Prototype → engineering handoff gap”

This issue is even more widely discussed.

The pattern:

- AI tools generate impressive prototypes quickly
- engineering teams then struggle to productionize them
- prototypes lack architecture, data models, scalability, edge cases, security, accessibility, testing, etc.

Evidence A — “Prototype-to-Production Gap”

[Why AI-Generated UI Fails in Production](#)

This article explicitly calls it:

“the prototype-to-production gap”

It explains that AI tools optimize for:

- visual polish
- speed
- demoability

but not:

- maintainable architecture
- engineering constraints
- production readiness

Instead of using a systematic approach, AI generates hardcoded values. Design tokens are standardized, named values for foundational design properties like colors, spacing, and typography. You'll see `#333333` instead of `color-text-primary`. Why does this matter? Future updates become painful. Theme changes require hunting through every file. Visual consistency at scale becomes nearly impossible to maintain. What seemed fast at first slows everything down later.

[Ministry of Programming](#) is a software product development company that helps startups and businesses design, build, and scale digital products, including web apps, mobile apps, and AI-powered platforms. They provide end-to-end services covering product strategy, UI/UX design, and software engineering.

[Replit](#) is a cloud-based coding platform that lets developers write, run, collaborate on, and deploy software directly from the browser. It supports multiple programming languages and includes AI-assisted development tools for faster app creation.

Additional Supporting Evidence

The role of generative AI in software development: Will it replace developers?

Failure to Keep to the Requirements - It was observed that across GitHub's Octoverse analysis of 1287 AI-generated code samples found that while 96.7% of code was syntactically correct, 73.4% did not capture the business requirements that were given.

Risk of Security Vulnerabilities - Stack Overflow's Developer Survey examined 4821 AI-suggested code implementations and found that 42.7% contained potential security flaws, including a 61.3% vulnerability rate in authentication, 44.8% in data validation and 39.2% in access control.

The core technical challenge: The context window

Difficulty Keeping Context in the Middle of Sessions - LLMs have a strong ability to recall information at the very beginning or end of a long text window, but suffer in information located in the middle. Which implies that requirements added on further into a session would likely be 'forgotten' by the AI.